

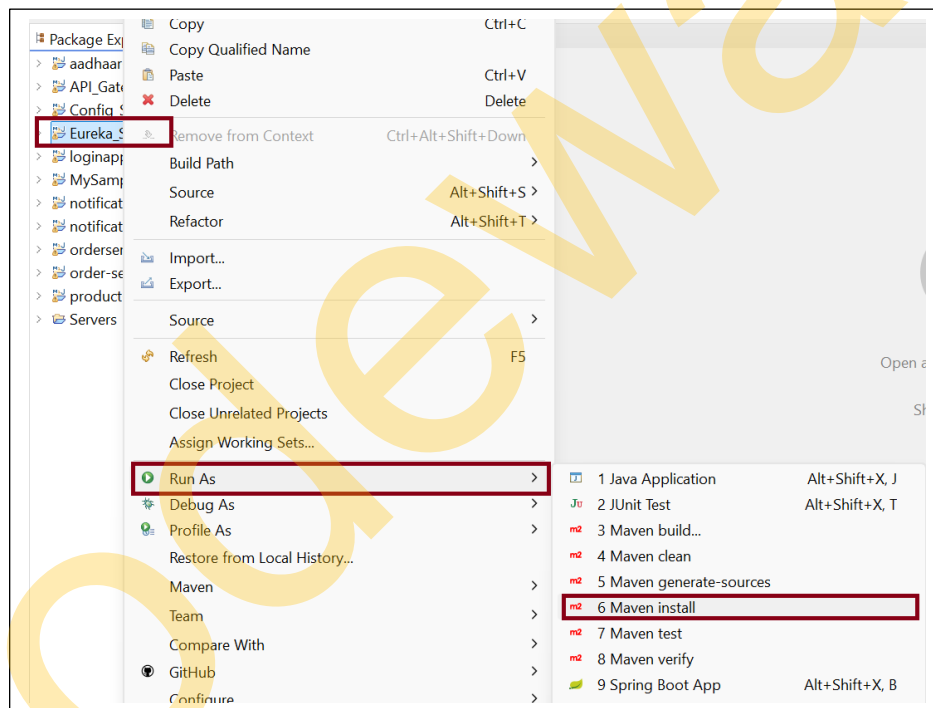
Deploying Spring Boot Microservices to Docker

1. Prerequisites

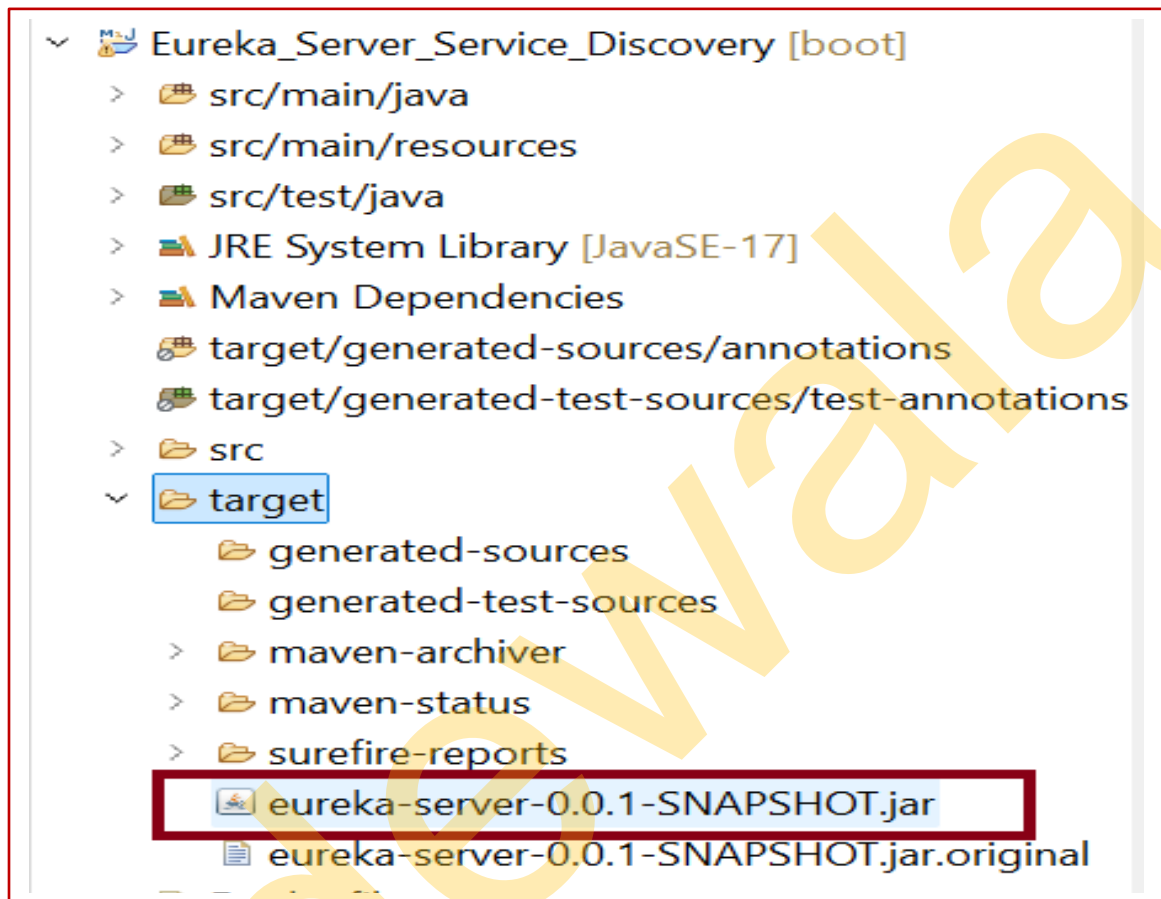
Before starting, make sure you have:

- Docker Desktop installed and running
- JDK 17+ installed
- Maven installed
- A working Spring Boot project(order service, product service etc).

2. Build the Spring Boot JAR



3. Once maven install is successful, it will generate jar file under “target directory”



4. Create a Dockerfile

Create a file named **Dockerfile** (no extension) in the project root (same folder as pom.xml)

Dockerfile example(just change **your application** jar name)

```
# 1. Use an official lightweight JDK base image
FROM openjdk:17-jdk-slim

# 2. Set working directory inside the container
WORKDIR /app

# 3. Copy the JAR file from target/ into container
COPY target/yourapp-0.0.1-SNAPSHOT.jar app.jar

# 4. Expose the port your app runs on
EXPOSE 8761

# 5. Command to run your application
ENTRYPOINT ["java", "-jar", "app.jar"]
```

5. Build the Docker Image

Go to the folder where docker file is created (service/project root folder) and run the docker build from cmd.

Command: `docker build -t eureka-server .`

```
C:\Users\Suresh\OneDrive\Desktop\Academy\microservice\Eureka_Server_Service_Discovery>docker build -t eureka-server .
[+] Building 46.0s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 384B
=> [internal] load metadata for docker.io/library/openjdk:17-jdk-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/3] FROM docker.io/library/openjdk:17-jdk-slim@sha256:aaa3b3cb27e3e520b8f116863d0580c438ed55ecfa0bc126b41f68c3f62f9774
=> => resolve docker.io/library/openjdk:17-jdk-slim@sha256:aaa3b3cb27e3e520b8f116863d0580c438ed55ecfa0bc126b41f68c3f62f9774
=> => sha256:6ce99fdf16e86bd02f6ad66a0e1334878528b5a4b5487850a76e0c08a7a27d56 187.90MB / 187.90MB
=> => sha256:44d3aa8d076675d49d85180b0ced9daef210fe4fdff4b422b9cf384e591d0 1.58MB / 1.58MB
=> => sha256:1fe172e4850f03bb45d41a20174112bc119fbfec42a650edbbd8491aee32e3c3 31.38MB / 31.38MB
=> => extracting sha256:1fe172e4850f03bb45d41a20174112bc119fbfec42a650edbbd8491aee32e3c3
=> => extracting sha256:44d3aa8d076675d49d85180b0ced9daef210fe4fdff4b422b9cf384e591d0
=> => extracting sha256:6ce99fdf16e86bd02f6ad66a0e1334878528b5a4b5487850a76e0c08a7a27d56
=> [internal] load build context
=> => transferring context: 58.25MB
=> [2/3] WORKDIR /app
=> [3/3] COPY target/eureka-server-0.0.1-SNAPSHOT.jar app.jar
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:b4041357f9ac4302ab882b4ba17a2287bae72e6a282d8424e830ef2b44dea607
=> => exporting config sha256:d427d2323ebb865aef8189d9cc9864e96ed7acb7ac595c4da2eba0f041974cc1
=> => exporting attestation manifest sha256:a219c71bf01270b77ff73e37fb5f054f8834cfff90ebfd7c6c3c360a407e4762ee
=> => exporting manifest list sha256:4c33e64f8a9fcef18d429b65e351145df6d300008d84c6a4b85855bdc94d0d2d
=> => naming to docker.io/library/eureka-server:latest
=> => unpacking to docker.io/library/eureka-server:latest
```

A new Docker image named eureka-server will be created.

You can verify using: `docker images` command.

```
C:\Users\Suresh\OneDrive\Desktop\Academy\microservice\Eureka_Server_Service_Discovery>docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
eureka-server       latest         4c33e64f8a9f   51 seconds ago  752MB
confluentinc/cp-kafka latest        9026dbbf280d   2 weeks ago    856MB
openzipkin/zipkin   latest        bb570eb45c29   5 months ago   379MB
confluentinc/cp-zookeeper latest       7610a50b13e7   6 months ago   1.75GB
hello-world         latest        940c619fbd41   9 months ago   20.4kB
confluentinc/cp-kafka 7.5.0         fbbb6fa11b25   2 years ago    1.33GB
confluentinc/cp-zookeeper 7.5.0        02f6c042bb9a   2 years ago    1.33GB
```

6. Run the Container

Command: `docker run -d -p 8761:8761 --name eureka-service eureka-server`

Explanation:

- d – Detached mode (runs in background)
- p – Maps local port 8761 to container port 8761
- name – Assigns name to container

```
C:\Users\Suresh\OneDrive\Desktop\Academy\microservice\Eureka_Server_Service_Discovery>docker run -d -p 8080:8080 --eureka-server cad33fc82c19ccdf8a87571334b9018dc849afee714f7fcd6d07283aeb82611
```

A container will start, running your Spring Boot Eureka Server.

Command: `docker ps`

7. Access the Application

Visit `http://localhost:8761` in browser to open Eureka dashboard.

8. View Logs

Command: `docker logs -f eureka-service`

Use -f to follow logs in real time.

9. Stop and Remove Container

Commands:

`docker stop eureka-service`

`docker rm eureka-service`

Happy Dockering! Keep calm and containerize on!